M.TECH

Roll No. | | | | | | | |

Paper Code: MSE 101

M.Tech (Structures)

Odd Mid Semester Examination 2018

I Semester

Paper Name :Theory of Elasticity and Plasticity

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks for each main question are ten.

Q1. **(5 X 2 =10 Marks)**

a. Describe the ideology behind various assumptions made in the theory of elasticity? (5)

b.The state of stress at a point with respect to 'xyz' Cartesian coordinate system were given as $\begin{bmatrix} 15 & 10 & -10 \\ 10 & 10 & 0 \\ -10 & 0 & 40 \end{bmatrix}$ Determine the magnitude of Normal and shear stress on a surface intersecting the point parallel to the plane given by 5x-2y+3z=0? (5)

c. Derive the Stress Transformation equation? (5)

Section B

Q2. **(5X2= 10 Marks)**

a. Write the steps for Mohr's Circle construction and how it is useful for Principal strain determination? (5)

b. Derive the equilibrium equations for plane strain condition? (5)

c. The state of stress at a point is given by $\begin{bmatrix} 9 & 6 & 5 \\ 6 & 5 & 2 \\ 5 & 2 & 4 \end{bmatrix}$ determine the principal stresses and principal directions? (5)

Q3. **(5X2= 10 Marks)**

a.Explain your understanding on Octahedral stresses? (5)

b.Principal Stresses for Plane Strain condition (5)

c.Hooks Law and Linear relations between the components of stress and the components of strain? (5)

Q4. (5 X 2 = 10 Marks)

a. Write the equations for principal stresses and Stress invariants? (5)

b. If the displacement field in a body is specified byu $= (2x^2 + 3)10^3$, $v = (3y^2z)10^3$ and $w = (2x + 3z)10^3$, determine the strain components at a point whose coordinates are (3,2,3)? (5)

c.Derive the equation of compatibility for strains? (5)

Q5. (5 X 2 = 10 Marks)

a.Explain the airy stress fuction? (3)

b.Write the stress components on a rectangular parallelepiped in Cartesian coordinate system (5)

c. Derive the equation of compatibility for stresses? (5)

4/oct
9:15

M.TECH

Roll No. | | | | | | | |

Paper Code: MSE-101

M.Tech. (Structural Engineering)
Mid Semester Examination 2018
1st Semester

Paper Name: Theory of elasticity and plasticity

Time: 1:30 Hours

MM: 50

Note : (i) This question paper contains two sections.
(ii) Section 'A' is compulsory
(iii) Answer any two questions from Section 'B'
(iv) Each question contains three parts.

Section A (Compulsory)

Q1. (20 Marks)

a. Briefly describe stress tensor. (7)
b. Explain Airy's stress function. (7)
c. What is generalized Hooke's law? (6)

Section B

Q2. (5 X 3 = 15 Marks)

a. Derive the equations of equilibrium in Cartesian co-ordinates. (5)
b. What is the specification of stress at a point? (5)
c. Explain St. Venant's principle. (5)

Q3. (5 X 3 = 15 Marks)

a. The stress components at a point are:

$$\sigma_x = 25Mpa \quad \sigma_y = 12Mpa \quad \sigma_z = 5Mpa$$
$$\tau_{xy} = 3Mpa \quad \tau_{yz} = 10Mpa \quad \tau_{zx} = 5Mpa$$

Determine the stress invariants. (5)

b. The principle stresses for the matrix given in 3(a) (5)
c. The direction cosines for the principle planes for the matrix given in 3(a) (5)

Q4. (5 X 3 = 15 Marks)

a. Write a short note on stress invariants. (5)
b. Show that stress is a second order tensor. (5)
c. What are compatibility equations? How are they useful? (5)

M. TECH

Roll NO. | | | | | | | |

Paper Code: MSE 102

4/oct/18
(3:30-5:00)

M. Tech
Mid-Semester Examination September/ October 2018
I semester
FEM in Structural Engineering

Time: 1:30 hrs

MM: 50

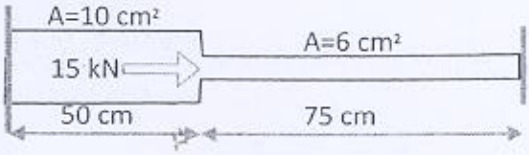
Note: This question paper contains two sections. Section 'A' is compulsory. Answer any two from section B

Section A (Compulsory)

Q1. (20 marks)

Based on the bar shown in the figure 1, answer questions a) , b) and c)

Figure 1

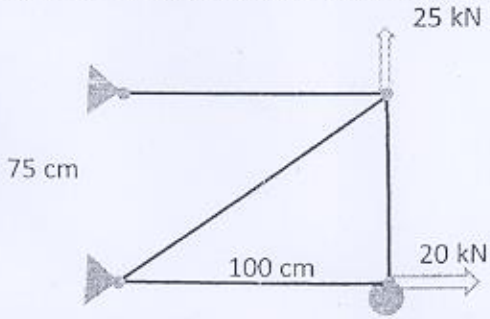


a) Derive the stiffness matrix for the bar shown in the figure 1 ? (7)
b) Obtain the nodal displacements, especially at the junction, for the bar shown above (7)
c) Determine the reactions at the two ends of the bar. (6)

Section B (Answer any TWO)

Q2. (5 x 3 = 15 marks)

a) Derive the element stiffness matrix for a 2-noded truss element using transformation(5)
b) Obtain the stiffness matrix for the truss shown below (E=20000 N/mm² and A = 60 mm² for all bars) (5)



c) Obtain the stress-strain relationship in matrix form, for three-dimensional case. (5)

Q3. (5 x 3 = 15 marks)

a) Using equilibrium equations obtain the relationship between the body forces and stresses in a 3D elemental volume dv (5)
b) Give a brief note on each of the following (i) Finite difference method (ii) Vibrational method, and (iii) Weighted Residual method (5)
c) Using linear shape functions for a 2-noded bar element, obtain the body force vector and traction force vector. Assume appropriate symbols (5)

Q4. (5 x 3 = 15 marks)

a) Derive the shape functions satisfying a C1 continuity requirement for a 2-noded Euler-Bernouli beam element (5)
b) Derive the 4x4 stiffness matrix for a two-noded Euler-Bernouli beam element (5)
c) Derive the consistent nodal force vector for uniform load on a two-noded Euler-Bernouli beam element (5)

Roll No. | | | | | | | |

M.Tech. (Structure)

Paper Code: MSE-105

(APPLIED MATHEMATICS)
Mid Semester Examination Oct, 2018
1st Semester

Time: 1:30 Hours MM: 50

Note:

(i) This question paper contains two sections.
(ii) Section 'A' is compulsory.
(iii) Answer any two parts of each question from Section 'B'.
(iv) Each question contains three parts.

Section A (Compulsory)

Q1 **(20 Marks)**

a. Find the greatest eigen value and corresponding eigen vector the following matrices using power method

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix}$$

(a) Solve the following system of equations by using Cholesky's method.

$$\begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

(b) Discuss the nature of the critical point (0, 0) for the linear system $\dot{X} = AX$.

Section B

Q2. **Attempt any two parts of choice from (a), (b), and (c)** **(5 x 2 = 10 Marks)**

a. Using Choleski's method, find inverse of the matrix $A = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$.

b. Solve the following system of linear equation by using by Gauss-Seidel's method.

$$x+y-z=0$$
$$-x+3y=2$$
$$x-2z=3.$$

Assuming the initial solution $X=(0.8,\ 0.8,\ 2.1)^T$

c. Discuss the all cases of phase plane method for a system $\dot{X}=AX$.

Q3. Attempt any two parts of choice from (a), (b), and (c) **(5 x 2 = 10 Marks)**

a. Prove that for the matrix $A=\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{bmatrix}$, all its eigen values are distinct and real

b. Test the stability of the following system of equations: $\dot{x}=3x+4y;\ \dot{y}=2x+y$

c. Solve the following system of linear equation by using by Jacobi method.

$$27x+6y-z=85$$
$$6x+15y+2z=72$$
$$x+y+54z=110.$$

Q4. Attempt any two parts of choice from (a), (b), and (c) **(5 x 2 = 10 Marks)**

a. Write LU decomposition Matrices by Crout's method for the given matrix

$$A=\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$$

b. What type of critical point does the following system of equations have?

$$\frac{dx}{dt}=x-y;\ \frac{dx}{dt}=2x+3y$$

c. Solve the following system of linear equation by using by Gauss's elimination method.

$$2x-y+3z=9$$
$$x+y+z=6$$
$$x-y+z=2.$$

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M.TECH

Roll No. | | | | | | | |

Paper Code: MSE 104

M.Tech (Structural Engineering)
Mid Semester Examination 2018
IstSemester

Computer Aided Analysis of Structures

Time: 1:30 Hours

MM: 50

Note :(i) This question paper contains two sections.
(ii) Section 'A' is compulsory
(iii) Answer any two questions from Section 'B'
(iv) Each question contains three parts.

Section A (Compulsory)

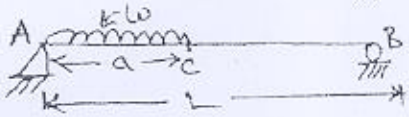
Q1.

(20 Marks)

a.What are the different types of loads and boundary conditions imposed on various structural members? (7)

b.Explain the conditioning of matrices with suitable example? (7)

c. Find the reaction force at end supports A and B? (6)



Section B

Q2.

(5X3= 15 Marks)

a.Factorize the given matrix using Crout's method (5)

$$A = \begin{bmatrix} 0 & 8 & 2 \\ 3 & 5 & 2 \\ 6 & 2 & 8 \end{bmatrix}$$

b.Solve the given linear algebraic equation using back substitution (5)

AX=B where $A = \begin{bmatrix} 0 & 8 & 2 \\ 3 & 5 & 2 \\ 6 & 2 & 8 \end{bmatrix}$ $B = \begin{bmatrix} -7 \\ 8 \\ 26 \end{bmatrix}$ and $X = \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}$

c.What do you mean by strictly diagonally dominant matrix? Check whether the matrix C is strictly diagonally dominant? (5)

$$C = \begin{bmatrix} 12 & 6 & 0 \\ 2 & -3 & 2 \\ 0 & 6 & 12 \end{bmatrix}$$

Q3.

(5X3= 15 Marks)

a.Find all the norms for given matrix? (5)

$$C = \begin{bmatrix} 12 & 6 & 0 \\ 2 & -3 & 2 \\ 0 & 6 & 12 \end{bmatrix}$$

b.Solve the set of linear equations using Jacobi iteration upto 3 iterations?(5)

$$5x-2y+3z=-1$$
$$-3x+9y+z=2$$
$$2x-y-7z=3$$

c.Solve the set of equations given in Ques. 3(b) using Gauss-Seidel Iteration method?(5)

Q4. **(5 X 3 = 15 Marks)**

a.Write the algorithm for LU decomposition using Doo-Little method?(5)

b.Explain the process of idealization and numerical modelling of structure? (5)

c.What are the assumptions for Jacobi iteration? (5)

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Roll No. | | | | | | | |

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M.Tech (Structures)

Odd Mid Semester Examination 2018

I Semester

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Q5.

a.Explain the airy stress fuction? (5)

b.Write the stress components on a rectangular parallelepiped in Cartesian coordinate system (5)

c. Derive the equation of compatibility for stresses? (5)